

Clinical Risk Assessment Report: Patient 82

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 41.3% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.87x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +5.7%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT)	Value: -0.31 [Bottom 21%] (+5.7%)
<i>Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.</i>	
Small Zone Emphasis (PET) (GLZLM_SZE.PET)	Value: 0.33 [Avg (64th %ile)] (+3.2%)
<i>Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.</i>	
75th Percentile Tumor Density (CONVENTIONAL_HUQ3)	Value: 0.71 [Avg (70th %ile)] (+3.1%)
<i>Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.</i>	
Squamous Cell Carcinoma Antigen (SCC_Ag)	Value: 0.6 [Bottom 18%] (+2.6%)
<i>Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.</i>	
Maximum Tumor Density (CONVENTIONAL_HUmax)	Value: 0.83 [Top 11%] (+2.3%)
<i>Reflects the most dense solid components or calcifications within the primary tumor lesion.</i>	
Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT)	Value: -0.21 [Bottom 16%] (+1.9%)
<i>Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.</i>	
Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT)	Value: -0.07 [Top 15%] (+1.6%)
<i>Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.</i>	
High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT)	Value: 0.68 [Avg (72th %ile)] (+1.2%)
<i>Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.</i>	

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

